Project Workflow: Week 1

Phase: Concept Design & Technical Planning

1. Objective The primary goal of Week 1 was to establish a solid creative and technical foundation for the Treasure Hunt browser game. This included:

- Defining the core gameplay loop and player objectives.
- Selecting a lightweight, dependency-free technology stack suited for browser-based 2D rendering.
- Conceptualizing the island world, exploration mechanics, and visual identity of the game.

2. Detailed Task Breakdown

- **Feature Identification:** Researched browser-based exploration and puzzle games to finalize the core gameplay pillars: procedural island generation, scroll-based riddle collection, torch fuel management, and a real-time scoring system based on discovery speed.
- **Technical Stack Selection:** Selected vanilla JavaScript (ES6+) paired with the HTML5 Canvas API for all rendering, avoiding external frameworks to keep the project lightweight, dependency-free, and easy to run with zero build steps. CSS3 was chosen for UI overlays, menus, and HUD styling.
- **Architectural Setup:** Designed a modular file architecture separating concerns cleanly: a core game loop (game.js), independent components for the map, player, and items, a dedicated rendering pipeline (renderer.js), and shared utility modules for math/randomization and persistent storage.

3. Deliverables

- **Project Scope Document:** A formal outline of the game's core mechanics and limitations.
- **Tech Stack Definition:** Finalized choice of vanilla JS, HTML5 Canvas, and CSS3.
- **Project Timeline:** A structured 10-week development roadmap.

4. Tools & Resources

- **Development Environment:** Node.js (for local dev server and live reload via npm run dev) and VS Code / any modern code editor.
- **Reference Materials:** MDN Web Docs — HTML5 Canvas API and vanilla JavaScript (ES6+) references.